PROJECT OVERVIEW

EnergyOS

Management System for Renewable Energy Operations — Monitoring, Control & Security for Grids, Substations, and Power Plants

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Audience: Grid Administrators, Substation Managers & Field Operators

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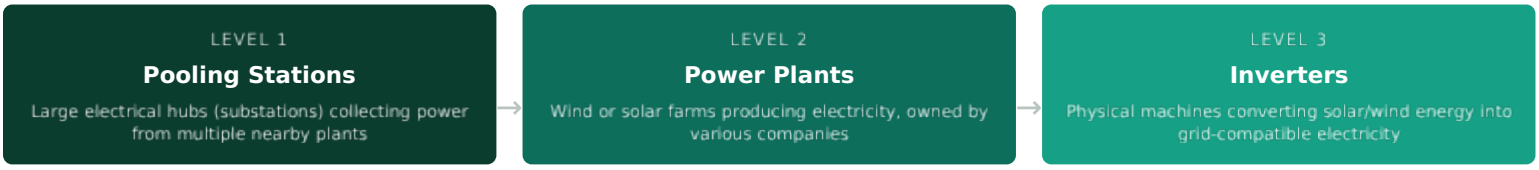
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1. Core Concept & Vision

In modern energy grids, keeping track of electricity generation and technical tuning parameters from solar and wind farms is critical for grid safety. EnergyOS acts as a unified digital control room, organizing grid infrastructure into a clear three-level hierarchy.



EnergyOS ensures that grid administrators, substation managers, and field operators all have their own tailored dashboards to monitor and update settings without stepping on each other's toes.

2. Who Uses the System?

EnergyOS employs a smart security setup to ensure users only see and edit what is relevant to their job.

Grid Administrators

The master users. They have complete visibility across all pooling stations and power plants. They manage user accounts, assign permissions, create new plants, and view the global activity logs.

Field Operators

The engineers on-site at the power plants. Based on their assignments, they can be:

- **Read-Only Operators** — View generation capacity and check if equipment is working.
- **Write-Enabled Operators** — Update equipment status (e.g., setting an inverter to "Under Maintenance" or "Online") and edit performance values.

Substation Managers (Pooling Station Users)

Local heads of the high-voltage electrical hubs. They have a dedicated dashboard showing data only for the power plants connected directly to their substation. They can monitor and update compliance configurations (Power Plant Controllers) for these plants.

3. Key Features Explained

A. Secure Sign-In with Email Verification (2FA)

To protect critical power grid configurations, logging into the system requires two steps:

- **Password Verification** — The user enters their email and password.
- **One-Time Code (2FA)** — A secure 6-digit code is instantly emailed to the user, who must enter it to access their dashboard.

Note: A bypass is enabled for official test accounts to speed up development checkups.

B. Power Plant Controller (PPC) Management

Every renewable plant has a master controller (PPC) that regulates voltage and grid frequency. EnergyOS allows managers to track and adjust these critical parameters:

- **Voltage & Power Settings** — Monitor control modes, voltage tolerances, and limits for reactive power injection or absorption.
- **Frequency Safety** — Track how plants reduce output during grid frequency spikes to prevent blackouts.
- **Communication Latency** — Track how fast controllers respond to grid instructions (in milliseconds).
- **CSV Exports** — Allows substation managers to download all controller configurations in spreadsheet (CSV) format for reporting.

C. Inverter Logs & Telemetry Registry

Tracks the hundreds of generating units at each power plant in a clean grid layout.

- **Equipment Status Updates** — Operators can switch an inverter's status between "Online," "Offline," "Fault," or "Maintenance" with a click.
- **Search & Filter** — Search for specific inverters by manufacturer, model, or status, and filter logs by power plant.
- **High/Low Voltage Ride-Through** — Monitor how inverters react to sudden grid voltage surges or drops.

D. Smart Permission Panel

Admins can customize which plants an operator is allowed to manage.

- **Dynamic Checks** — The system prevents mistakes; granting Write access automatically checks Read access too.
- **Batch Saving** — Admins can configure multiple plant permissions for a user and save them all at once rather than one plant at a time.

E. Grid Frequency Analytics

A specialized view that groups generating equipment based on how they react to sudden power grid frequency disturbances. This helps grid engineers verify if there is enough reserve capacity to stabilize the grid during an emergency.

F. Security Audit Ledger (Activity Logs)

A permanent diary of all changes made to the system. Whenever a user edits a plant, changes a controller config, or registers a new user, the system automatically logs:

- Who made the change (username)
- When it occurred
- What exactly was changed (e.g., showing the old value alongside the new value)

4. Key Benefits of EnergyOS

Enhanced Security
Protects critical grid infrastructure through 2-Factor Authentication and granular access limits.

Improved Coordination
Connects grid coordinators, substation managers, and field developers in one collaborative space.

Regulatory Compliance
Log ledger ensures all settings adjustments are audited and traceable, meeting national load dispatch requirements.

User-Friendly Dashboards
Complicated technical data is visualized through clean graphs, charts, and table lists.